ISD 521
Project
Presentation

Team 17: Mechatronics and Embedded Systems
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Stakeholder Persona and Needs



Our Persona is a senior research scientist named Dr. Emma Wright. Dr. Wright leads a research team at a prominent university lab focused on genetic research and development of new therapies.

- involves repetitive and precise tasks such as pipetting, sample preparation, and analysis
- a focus on efficiency and throughput.

Her pain points lead back to the repetitive tasks involving pipetting and moving tubes/plates for sample preparation or downstream analysis

- Time-consuming
- cause physical strain
- take away much needed manpower
- often suffer from human errors

Dr. Wright also expresses a strong interest towards **advanced technology** in research innovation especially if they show benefits in efficiency and throughput while **balancing budget constraints**.

Translation to Requirements

- Lab Productivity (“I want the solution to improve lab productivity”)
- Solution Performance (“I want the solution to perform to a high standard”)
- Solution Integration (“I want the solution to work with existing tool processes and personnel”)
- Safety and Comfort (“The solution should be safe to work with and pleasant to work alongside”)
- Solution Operation (“I want the solution to be easy to operate”)
- Maintenance and Lifetime (“The solution should require minimal maintenance and perform well over its lifetime”)

Expanded Requirements

#	Name	Text
1	1 Lab Productivity	I want the solution to improve lab productivity
2	1.1 Human Error	Minimize human error in repetitive tasks.
3	1.2 Menial Labor	Allow researchers to use their time for more complex, high-value tasks.
4	1.3 Lab Efficiency	Increase lab efficiency and throughput.
5	1.4 Precise Handling	Ensure consistent and precise sample handling to improve data reliability.
6	1.5 Technology Integration	Integrate advanced technology to stay at the forefront of research innovation.
7	1.6 Cost-Effectiveness	It's important that the unit is cost-effective in the long run. We're looking for something that saves us money on labor and helps us increase productivity.
8	2 Solution Performance	I want the solution to perform to a high standard
9	2.1 Versatility	The unit needs to be able to be easily reconfigured between tasks.
10	2.2 Flexibility:	The unit should be able to switch between different tasks—like pipetting, mixing, and moving samples—without a lot of hassle. We want it to adapt to our varying protocols easily.
11	2.3 Data Management:	We need the unit to log data and track samples efficiently. It should integrate well with our lab information management systems so we can keep everything organized.
12	2.4 Precision and Accuracy:	We need the unit to be precise, with an accuracy of ±0.1 mm, so that every sample is handled correctly and our results are reliable.
13	3 Solution Integration	I want the solution to work with existing tools processes and personnel
14	3.1 Compact Design:	The unit should have a compact footprint, allowing it to fit within standard laboratory spaces, including a biosafety cabinet, while maintaining full functionality.
15	3.2 Compliance with Standards	"The unit must meet relevant ISO and FDA standards, such as ISO 13485 and 21 CFR Part 820, to ensure our lab complies with industry regulations for quality management."
16	3.3 Integration with Existing Systems:	The unit should work smoothly with our current lab equipment and software. We want to integrate it into our existing workflow without needing to change everything.
17	3.4 Movable Weight	The unit should be light enough to be carried easily when moved around the lab.
18	4 Safety and Comfort	The solution should be safe to operate and pleasant to work alongside
19	4.1 Safety Features:	We want the unit to have emergency stop buttons and sensors to avoid accidents, so our team can work confidently around it.
20	4.2 Incident Reporting	"It must have a mechanism for reporting and documenting any incidents or deviations during operation, so we can comply with regulatory reporting requirements."
21	4.3 Safety Certifications	"It should have appropriate safety certifications, such as CE marking or UL certification, to confirm that it meets safety and environmental standards."
22	4.4 Noise Level Compliance	"The unit must operate at a sound level of no more than 60 dB to ensure a comfortable working environment for our staff, complying with occupational noise regulations."
23	4.5 Chemical Handling Compliance	"The unit must be designed to handle chemicals safely, in accordance with OSHA regulations, to protect our staff and the environment."
24	5 Solution Operation	I want the solution to be easy to operate
25	5.1 User-Friendly Interface:	It has to have a simple control system. We don't want complicated programming; a touchscreen with a clear layout would be ideal so that our team can operate it without extensive training.
26	5.2 Calibration and Validation	"We require the unit to support routine calibration and validation processes to ensure ongoing compliance with quality standards and accurate performance."
27	5.3 Training and Competency Requirements	"The system should include training resources to ensure all users are competent in its operation, meeting our internal training requirements as well as regulatory expectations."
28	6 Maintenance and Lifetime	The solution should require minimal maintenance and perform well over its lifetime
29	6.1 Data Integrity	"The unit must ensure data integrity by complying with 21 CFR Part 11, so that all electronic records are accurate, secure, and reliable."
30	6.2 Maintenance Records	"We need a system that automatically logs maintenance activities and repairs to maintain compliance with regulatory requirements for equipment upkeep."
31	6.3 Maintenance and Support:	It should come with easy-to-follow maintenance guides, and we want quick access to customer support for any issues that might come up.
32	6.4 Documentation and Traceability	"We need comprehensive documentation for all processes and operations. The system should provide traceability for each task to facilitate audits and inspections."

Academic Version for Teaching Only
Commercial Development is strictly Prohibited

Measures of Effectiveness

Overarching Requirement	MOE	Justification for requirement
Lab Productivity	Time saved, the hours lab staff can now dedicate to other activities	Arguably the system's most important metric. As a system designed to reduce the workload of lab staff, if the system creates more work for the lab staff it will be considered a failure
Solution Performance	Placement accuracy, measured as incidents of error per 100 tasks completed	A measure of how well the system performs, useful for understanding the quality of the system as a whole
Solution Integration	Functional requirement, necessitates proper interfacing with other lab equipment and operator	The system will not be used in isolation, so it is important that it works well with existing workflows and equipment to support overall lab operations and objectives.
Safety and Comfort	Sound level, the sound disturbance the solution will create during operation	The system will operate in an environment with other people, so it shouldn't be irritating to work with or around.
Solution Operation	Functional requirement, subjective to the lab staff's perception of the ease of use of the solution.	The system will be used by lab staff, and should be intuitive to interact with and operate for maximum efficiency.
Maintenance and Lifetime	Weight, the solution weight in case it needs to be moved for maintenance and or repairs throughout its lifetime.	The system as a piece of the lab needs to be able to be moved around, so excessive weight should be avoided.

Technical Decisions: Solution Functionality

Solution will perform 2 main functions:

1. Pick and place
2. Pipetting

Both functions derived from stakeholder needs

Solution Brainstorming





Options that were considered:

- Liquid Handling Workstation
  - Designed for pipetting/mixing/dispensing with high precision and throughput using gantry
  - Large space requirements, high cost
- Pneumatic Pick-and-Place System
  - Use suction or grippers to move tubes, vials, or plates
- High-throughput Screening System
  - Customized for large-scale experiments, combining liquid handling and pick/place
  - Highly complex, expensive
- Manual Labor
  - Hire external staff to perform lab activities
- Robotic arm
  - Small footprint, able to integrate with existing equipment. Capable of both liquid handling and pick/place operations.
  - Flexible and easy to program, lower complexity and development time

Solution Selection

- Liquid Handling Workstation
  - High Cost, unable to pick/place
- Pneumatic Pick-and-Place System
  - Unable to do pipetting
- High-throughput Screening System
  - Complex and expensive
- Manual Labor
  - Human error, Expensive
- **Robotic arm**
  - Fulfills all functions, less expensive, easy to use and less complex

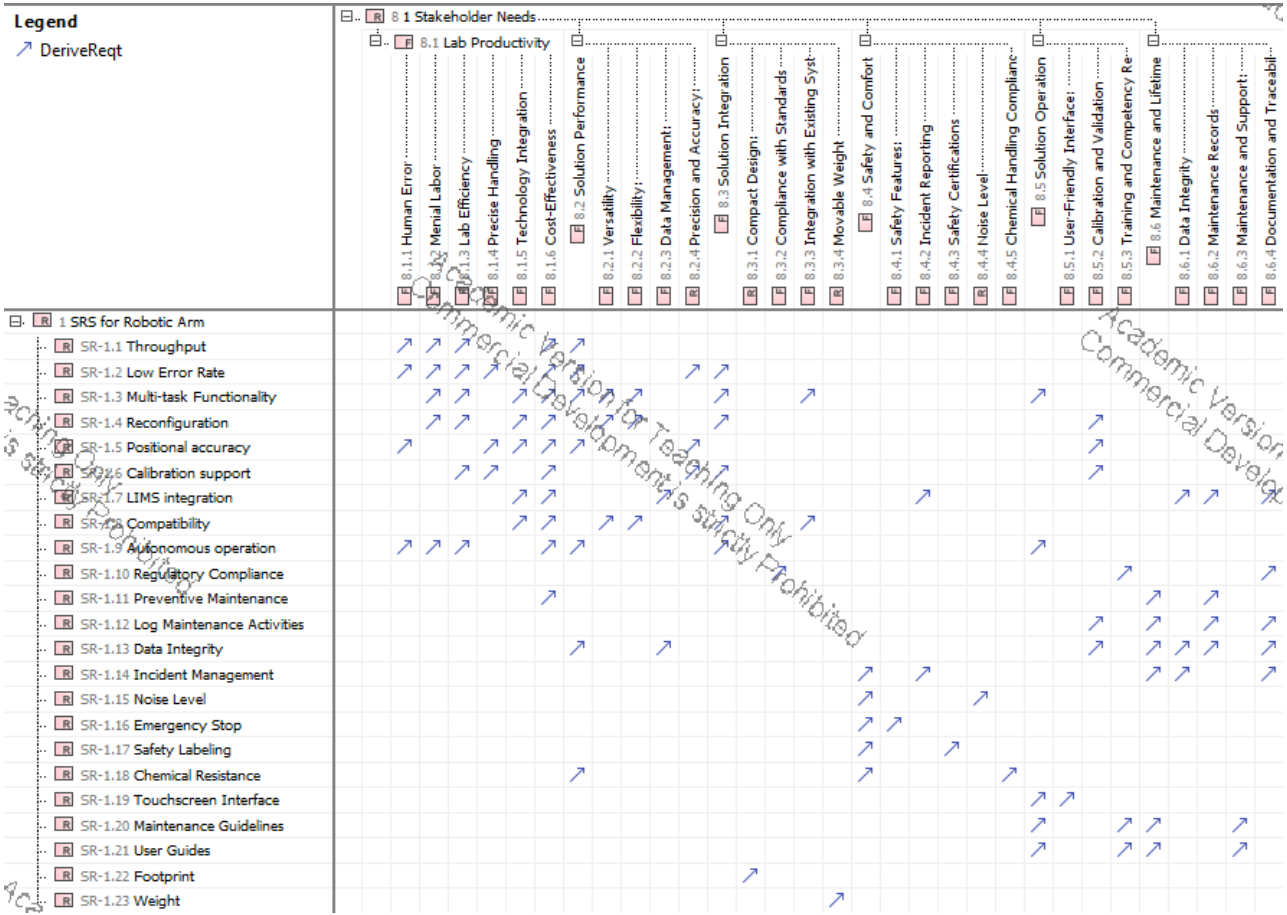




Technical Decisions: Subsystem Partitioning

- Power Management Subsystem
  - Reception, distribution, and transforming of power in the system.
- Manipulator Subsystem
  - Control and movement of samples, as well as taking required data on the samples as needed.
- Embedded Controller Subsystem
  - Controls the robot's movements and communicates with the lab equipment to receive and send information.
- Physical Architecture Subsystem
  - Physical resistance, weight, and stability of the robot's main body and mounting hardware.

Requirements Matrix- Fulfillment of all requirements



Verification Plan: How do we know a solution that meets requirements will satisfy the stakeholders

- Functional Requirements: Verification through performance of system via prototypes and software simulations.
- Non-functional Requirements: Verification through calculations of subsystem performance.
    - Ex) System noise requirements - added noises from each subsystem. Subsystem noises estimated based on research into comparable systems. Multiple values combined to be less than the 60dB requirement threshold.

Overall Team Process

- Project took around 50 hours to complete
- Work was done in two ways
  - Done together while on call with each other
  - Individually done based on time available
- Members could bring in technical experience
  - A team member has worked in a laboratory environment
  - A team member worked with robots similar to a robot arm

Thanks for listening!